

PENGYU JIE



Looking for a 4-6 months paid full-time internship from Feb 2026

Rouen, France

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Portfolio: <https://rajayoux.github.io> GitHub: <https://github.com/JayGarland>

Summary

MSc Software Engineering student specializing in XR/VR development (Unity/C#) and backend systems. Skilled in building interactive gameplay features, XR input pipelines, and modular systems, with additional experience in VR optimization from industry work at MangoTV. Also developing full-stack applications with .NET APIs, Python automation, Docker, and Azure. Currently building a Moto Tour booking system with complete backend workflow and API design.

Tech Stack

C# • Unity • .NET • Python • Docker • Azure • Angular • OpenXR • MRTK • SQL • REST API • GitHub Actions • CI/CD

Technical Experience & Projects

MangoTV, Changsha, China

Dec 2023 – Apr 2024

VR Applications & Rendering Intern

- Reduced scene loading time from 120s → 60s by implementing Unity Addressables + async texture streaming, profiling CPU/GPU stalls using Unity Profiler & GPU Frame Debugger.
- Enhanced runtime stability by redesigning asset loading logic with lazy initialization and environment-based resource management.

Reddit Auto Reply Bot

170★ on GitHub

- Designed a modular AI-reply pipeline supporting multiple models (Azure AI Inference, Gemini, Cohere, DeepSeek) via a provider-based architecture, allowing easy extension without modifying core logic.
- Built intelligent context analysis using conversation builders, subreddit-style learning, and user-history analyzers to produce context-aware replies.
- Containerized the entire application with Docker multi-stage builds, enabling portable deployment and lightweight images for Azure Container Apps.

MRPuzzle

Mixed Reality Puzzle (Quest 3, MRTK, OpenXR)

- Built natural hand-tracking and controller interactions using MRTK poke/grab solvers, enabling precise puzzle-piece manipulation across both input modes.
- Anchored the MR puzzle board to real surfaces using Oculus spatial anchors, ensuring stable alignment of virtual tiles with the player's physical table.

Education

ESIGELEC, Rouen, France

Sep 2024 – Jun 2026

MSc: Software Engineering & Digital Transformation

GPA 16.45/20

HVUST, Haikou, China

Sep 2020 – Jun 2024

B.S. Software Engineering

GPA 3.7/4

Awards

Malaysia Technology Expo (MTE 2023)

Gold + Special Award

China Collegiate Computing Contest (2022&2023&2024)

Provincial First Prize && National Second Prize

Provincial Excellent Undergraduate Innovation Training Program (2023)

Top 10%

Languages

English B2

French B1

Chinese Native

Hobbies

Hiking, Cycling, and Building Customized PC